

Should You Still Learn to Code in an A.I. World?

Coding boot camps once looked like the golden ticket to an economically secure future. But as that promise fades, what should you do? Keep learning, until further notice.



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By Sarah Kessler

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When Florencio Rendon was laid off from his third construction job in three years, he said, “it was the straw that broke the camel’s back.”

He was 36, a father of two, and felt time was running out to find a career that would offer higher pay and more stability. “I’ve always been doing jobs that require physical labor,” he remembers thinking. “What if I start using my brain for once?”

An Army veteran, Mr. Rendon explored training programs he could fund using his military benefits. He landed on a coding boot camp.

At first, the intensive courses seemed intimidating. Mr. Rendon had gotten his high school equivalency diploma before joining the Army, and he had taken some college courses, but he didn’t consider himself book smart.

Still, he thought about his children, who are now 4 and 2, and reasoned, “If I can make this work, then I should at least give it a try.”

His application to a course run by the company Fullstack Academy was accepted, and he started classes in April 2023, with a grant for military veterans that covered the \$13,000 tuition. While the material was challenging, he was pleasantly surprised to learn he could get the hang of it, and four months later, he graduated from an online program that he completed from his home in the Bronx.

The setback came after graduation: “Little did I know,” Mr. Rendon said of his new skills, “that’s not enough to get a job.”

Between the time Mr. Rendon applied for the coding boot camp and the time he graduated, what Mr. Rendon imagined as a “golden ticket” to a better life had expired. About 135,000 start-up and tech industry workers were laid off from their jobs, according to one count. At the same time, new artificial intelligence tools like ChatGPT, an online chatbot from OpenAI, which could be used as coding assistants, were quickly becoming mainstream, and the outlook for coding jobs was shifting.

Mr. Rendon says he didn’t land a single interview.

Coding boot camp graduates across the country are facing a similarly tough job market. In Philadelphia, Mal Durham, a lawyer who wanted to change careers, was about halfway through a part-time coding boot camp late last year when its organizers with the nonprofit Launchcode delivered disappointing news.

“They said: ‘Here is what the hiring metrics look like. Things are down. The number of opportunities is down,’” she said. “It was really disconcerting.”

In Boston, Dan Pickett, the founder of a boot camp called Launch Academy, decided in May to pause his courses indefinitely because his job placement rates, once as high as 90 percent, had dwindled to below 60 percent.

“I loved what we were doing,” he said. “We served the market. We changed a lot of lives. The team didn’t want that to turn sour.”

Compared with five years ago, the number of active job postings for software developers has dropped 56 percent, according to data compiled by CompTIA. For inexperienced developers, the plunge is an even worse 67 percent.

“I would say this is the worst environment for entry-level jobs in tech, period, that I’ve seen in 25 years,” said Venky Ganesan, a partner at the venture capital firm Menlo Ventures.

For years, the career advice from everyone who mattered — the Apple chief executive Tim Cook, your mother — was “learn to code.” It felt like an immutable equation: Coding skills + hard work = job.

Now the math doesn’t look so simple.

Irresistible A.I.

Since their emergence in the mid-2010s, intensive courses in basic coding skills have been praised as a quick route to a high-paying career, especially for people who didn’t graduate from college. President Barack Obama made them part of his jobs initiative, nonprofits set them up to propel people of diverse backgrounds into tech careers, and universities from Harvard to Berkeley offered their own versions.

And they worked. In a 2020 survey of 3,000 boot camp graduates by CourseReport, 79 percent of respondents said the courses had helped them land a job in tech, with an average salary increase of 56 percent.

But the industry pulled back from hiring at the same time that new A.I. coding tools were starting to become mainstream. In 2022, Google’s A.I. team, DeepMind, reported that it had tested its A.I. model AlphaCode in coding

competitions, and that it was as good as “a novice programmer with a few months to a year of training.”

It took a few more years, but the tools available to a typical programmer have since improved markedly. This September, OpenAI released a new version of ChatGPT. It computes answers in a way that is different from previous models and may be even better at writing code. Tools like AlphaCode from Google and Copilot from GitHub generate snippets of code for specific purposes, testing or optimizing existing code and finding bugs.

The real proof is among developers: About 60 percent of 65,000 developers surveyed in May by StackOverflow, a software developer community, said they had used A.I. coding tools this year.

Not everyone sees these developments as a death knell for coding jobs. Armando Solar-Lezama, who, as the leader of M.I.T.'s Computer Assisted Programming Group, spends his days thinking about how to bring more automation into coding, said A.I. tools still lacked a lot of the essential skills of even junior programmers. His research has shown, for example, how large language models like GPT-4 failed to truly understand the problems they were solving with code and made sometimes ridiculous mistakes.

“When you’re talking about more foundational skills, knowing how to reason about a piece of code, knowing how to track down a bug across a large system, those are things that the current models really don’t know how to do,” he said.



Armando Solar-Lezama of M.I.T. finds that A.I. coding tools still lack some essential skills of even junior programmers. Veasey Conway for The New York Times

Still, A.I. is changing how software is made. In one study, an A.I. Coding assistant made developers 20 percent more productive. Google's chief executive, Sundar Pichai, said on a recent call with analysts that more than a quarter of the company's new code was now generated by A.I., but reviewed and accepted by engineers.

As with any discussion about automation, there are two ways people tend to forecast the outcomes of this development. Mr. Solar-Lezama believes that A.I. tools are good news for programming careers. If coding becomes easier, he argues, we'll just make more, better software. We'll use it to solve problems that wouldn't have been worth the hassle previously, and standards will skyrocket.

The other view: “I think it’s pretty grim,” said Zach Sims, a co-founder of Codecademy, an online coding tutorial company. He was talking specifically about the job prospects for coding boot camp graduates.

Hiring: GPT Monkeys

To be clear, both Mr. Solar-Lezama and Mr. Sims — and just about everyone working in technology whom I interviewed for this article — still think you should learn to code. But some see a parallel with long division: It’s good to understand how it works. It’s an arguably necessary exercise for learning more advanced mathematics. But on its own, it gets you only so far.

Matt Beane, an assistant professor of technology management at the University of California, Santa Barbara, is studying how the use of A.I. tools is already affecting entry-level coders at five large corporations across industries like banking and insurance.

“The phrase GPT monkey has come up repeatedly and independently,” he said. “They feel like they are relegated to small tasks that they just sort of churn through with the help of some A.I.-related tool.”

Sometimes, the new coders he is tracking don’t even get the opportunity to do that. Because A.I.-generated code is riddled with errors that are hard to spot without experience, senior developers sometimes find it easier to generate and edit it themselves than to let it fall to a junior programmer.

Mr. Beane observed the same conundrum with other skills in which work was being automated, like surgery and financial analysis: Beginners need more expertise to be useful, but getting the type of experience that would normally help build that expertise is becoming harder.

For a while, basic coding skills were a clear on-ramp to a tech career for people like Mr. Rendon who didn't have a college education or a lot of experience. In the future, entry-level coders may need a broader range of skills and more training to be effective. They may have to understand more about how their code works within a broader system.

Strategizing around business problems is also becoming more important, said Stephanie Wernick Barker, the president of Mondo, a tech staffing and recruitment firm, "So college degrees are still king."

In other words, the biggest change taking shape in software jobs may be not that A.I. replaces software engineers, but that it makes it more difficult to become one.

Stay Sharp. Keep Learning.

In the arena of cliché job advice, "learn to code" has been replaced by a call for "A.I. skills."

M.I.T., Cornell, Northwestern, Columbia and other universities now lend their names to A.I. certificates. Fullstack Academy, the coding boot camp Mr. Rendon attended, recently started a 26-week A.I. and machine learning boot camp. And companies like Booz Allen and JPMorgan Chase are offering free A.I. courses to employees.

The most popular job titles specific to A.I. include "machine-learning engineer" and "artificial intelligence engineer," according to CompTIA. Some skills listed in these job postings are "deploying and scaling machine-learning models" and "automating large language model training, versioning, monitoring and deployment processes."

You can't learn that quickly without a math or coding background.

Another category of “A.I. skill” feels more elusive. In a recent survey of more than 9,000 executives by Microsoft and LinkedIn, 66 percent said they wouldn’t hire someone without A.I. skills, but it’s unclear, exactly, what those skills look like.

It doesn’t help that the technology is moving quickly: Depending on whom you ask, we may be either a few years or many decades away from A.I. that can basically do anything the human brain can. When I asked Mr. Beane what we should be teaching young people to make them employable, he said: “You have to just stay sharp. You have to keep learning. Until further notice.”

Robert Wolcott, a venture investor who teaches business classes at both Northwestern’s Kellogg School of Management and the University of Chicago Booth School of Business, said he tells anxious parents that their children should study whatever they’re passionate about, even if it’s ancient architecture, but also take a class in statistics, accounting and computing.

“I think you learn to learn,” said Mr. Ganesan, the venture capitalist.

Mike Taylor, the chief technology officer of the global tech services company World Wide Technology, provided perhaps the most straightforward list: “problem solving skills,” “business acumen and values” and “clear and persuasive communication skills.”

Compared with “learn to code,” though, this is not easily actionable advice. For Mr. Rendon, the Fullstack Academy graduate, the dilemma isn’t an abstract one.

When he didn’t land any interviews for coding jobs, he went back to construction. The project finished, and he was laid off again.

When I first spoke with him in early August, he was pondering a choice. He was interviewing for a job with the Border Patrol, which would require moving his family out of New York. But he had also learned that his veterans’

benefits would provide him with enough housing assistance that he could go to college to study computer science.

College seemed like a good idea, “but what if I go this route and it doesn’t work out?” he asked.

Two months later, he had enrolled in the college classes. In his first computer science class, the professor went over the history of computers. It was a lot different from coding boot camp.

“This is more like general stuff that opens the possibility for other things,” he said.

Sarah Kessler is an editor for the DealBook newsletter and writes features on business and how workplaces are changing. [More about Sarah Kessler](#)